# Hamed Hojatian

*Ph.D. Candidate*

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## Education

Fall 2019 – Now
**Polytechnique Montréal**
*Ph.D. in AI-aided wireless communication systems*
**Supervisors:** Prof. François Leduc-Primeau, Prof. Jean-François Frigon
*GPA: 4.0/4.0*

2014-2017
**Isfahan University of Technology (IUT)**
*M.Sc. in communication system engineering*
**Supervisor:** Prof. Mohammad Javad Omidi
*GPA: 3.7/4.0*

2010-2014
**Babol (Noshirvani) University of technology (NIT)**
*B.Sc. in electrical engineering*
**Supervisor:** Prof. Hossein Miar-Naeimi
*GPA: 3.1/4.0*

## Research Experience

Ph.D.
***AI-aided Beamforming in massive MIMO Systems***
- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design and intelligent beamforming;
- Coordinated hybrid beamforming and cell-free massive MIMO with deep neural network; and
- Subarray hybrid beamforming and efficiecnt hardware design driven by unsupervised learning.
- ...

M.Sc.
***Time and Frequency synchronization in OFDM-based Massive MIMO systems***
- Time and frequency synchronization in OFDM-based massive MIMO;
- Carrier Frequency Offset (CFO) and channel estimation;
- Detection in receiver using iterative method; and
- Improving the performance by using Filtered-OFDM.

B.Sc.
***OFDM Transmitter and receiver simulation***
- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

## Skills

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|---|---|
| Tools and Languages | Python, Pytorch, TensorFlow, MATLAB, HTML and CSS, JavaScript, Bash, and Microsoft office. |
| Telecomm. Tools | Ericsson OSS, MapInfo, Atoll, TEMS, Actix |
| Languages | English, French (basic), Persian (native) |

## Community Service

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|---|---|
| IEEE Journals reviewer | Selected Areas in Communications (**J-SAC**), Transactions on Wireless Communications (**TWC**), Transactions on Communications (**TCOMM**), Wireless Communications Letters (**WCL**), Communications Letters (**CommL**), and Transactions on Vehicular Technology (**TVT**). Details in here. |

## Research Interests

- AI-aided wireless communication;
- Machine Learning, Deep Learning, DRL, Federated Learning, META and few shot learning;
- Statistical learning, optimization;
- Signal processing in massive MIMO communication, 5G beyond and 6G;
- Intelligent beamforming, coordinated beamforming, distributed antennas, cell free communication; and
- Millimeter wave in Massive MIMO and antenna array.

## Work Experience

Nov 2015 - Nov 2018

**Mobile Telecommunication Company of Iran (MCI)**
*Radio planning and optimization engineer*

- Live network Ericsson OSS optimization and parameter tuning for 2G, 3G, and LTE;
- Network Planning (2G, 3G and LTE), new site baseline parameters tuning and planning;
- Project manager of RNC rehoming and RNC planning; and
- Project manager of site renaming.

Oct 2015

**Eleventh ($11^{th}$) national robotics competition at BABOL University of Technology**
*Vice-President*

## Publications

2022

1. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning**". ***IEEE Communications Letters*** (2022).
2. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Flexible Unsupervised Learning for Massive MIMO Subarray Hybrid Beamforming**". ***IEEE GLOBECOM*** (2022).

2021

3. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming**". ***IEEE Transactions on Wireless Communications*** (2021).

2020

4. **Hojatian, H.**, Nguyen Ha, V., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**RSSI-Based Hybrid Beamforming Design with Deep Learning**". ***IEEE ICC*** (2020).

2016

5. **Hojatian, H.**, Omidi, M. J., Saeedi-Sourck, H. & Farhang, A. "**Joint CFO and Channel Estimation in OFDM-based Massive MIMO Systems**". ***IEEE IST*** (2016).

## Talks and Teaching Experience

2022 Hybrid-Seminars on Selected Topics in Signal Processing.
Presenting my research progress in **IVADO - Digital October.**

2021 Presenting my research progress in **IVADO - Digital October.**

2020 Paper Presentation, **IEEE ICC 2020**, Dublin, Ireland.
Presenting my research topic in **IVADO - Digital October.**

2013 Teaching assistant for **Electronic I**.
Teaching assistant for **Electronic circuit I**.

2012 Teaching assistant for **Digital Signal Processing**.

## Certificates

2022 Introduction to Unix Shell, **Calcul Quebec**.
Massive data analysis with Spark, **Calcul Quebec**.
Introduction to GPU Programming with Python, **Calcul Quebec**.

2019 $4^{th}$ **IVADO/MILA** SCHOOL IN DEEP LEARNING.

2018 DEEP LEARNING WITH PYTORCH, **Udemy**.
LTE OPTIMIZATION OF RADIO NETWROK (3GPP), **TechCom**.

2017 LTE COUNTER AND KPI (ERICSSON), **TechCom**.
LTE RADIO NETWORK PLANNING, **TechCom**.

## Mentoring

- A master student at Polytechnique Montréal.
- An internship research student from Cameron.

## Strengths

Hard-working, Decision Making, Ability to Work Under Pressure, Motivator & Leader Conflict Resolution, Time Management, Target oriented, Flexibility, and Adaptability